

Tiled Chip Multicore Processors



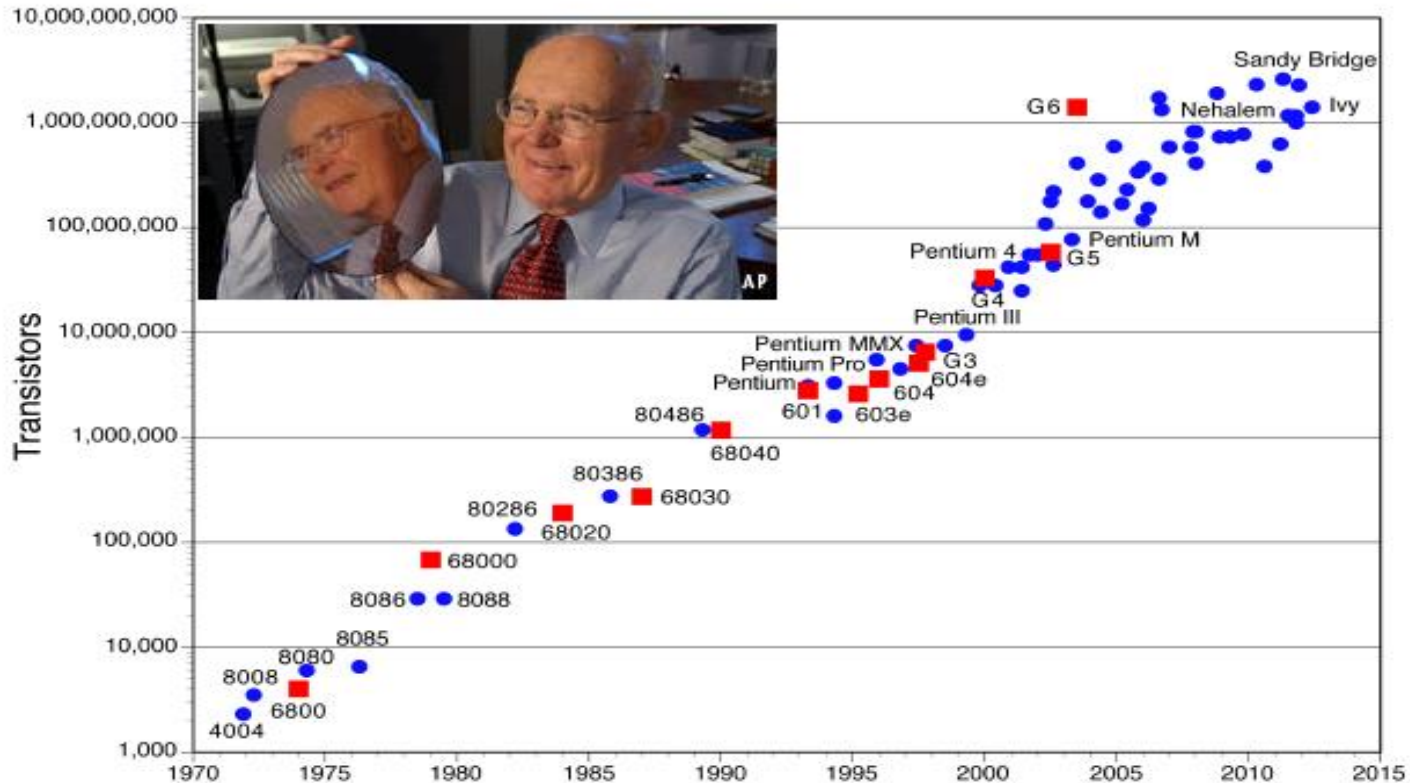
John Jose

Associate Professor

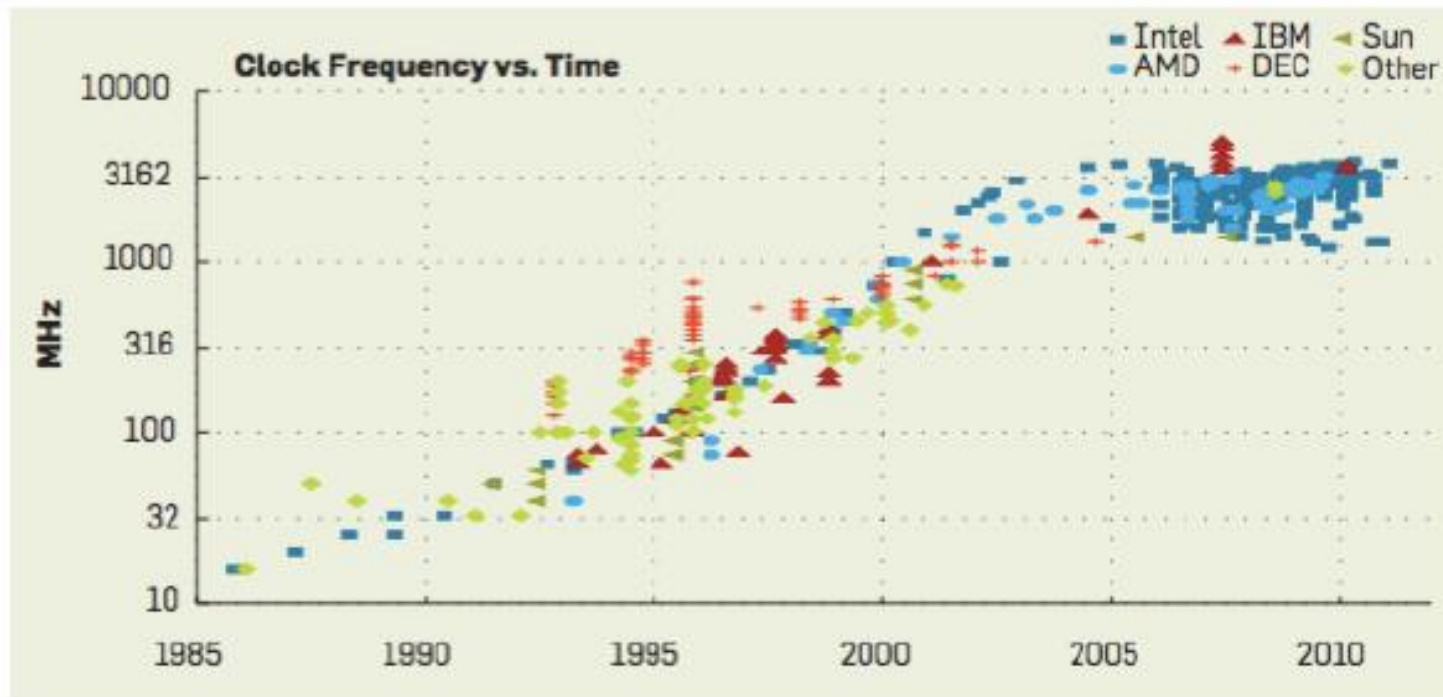
Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

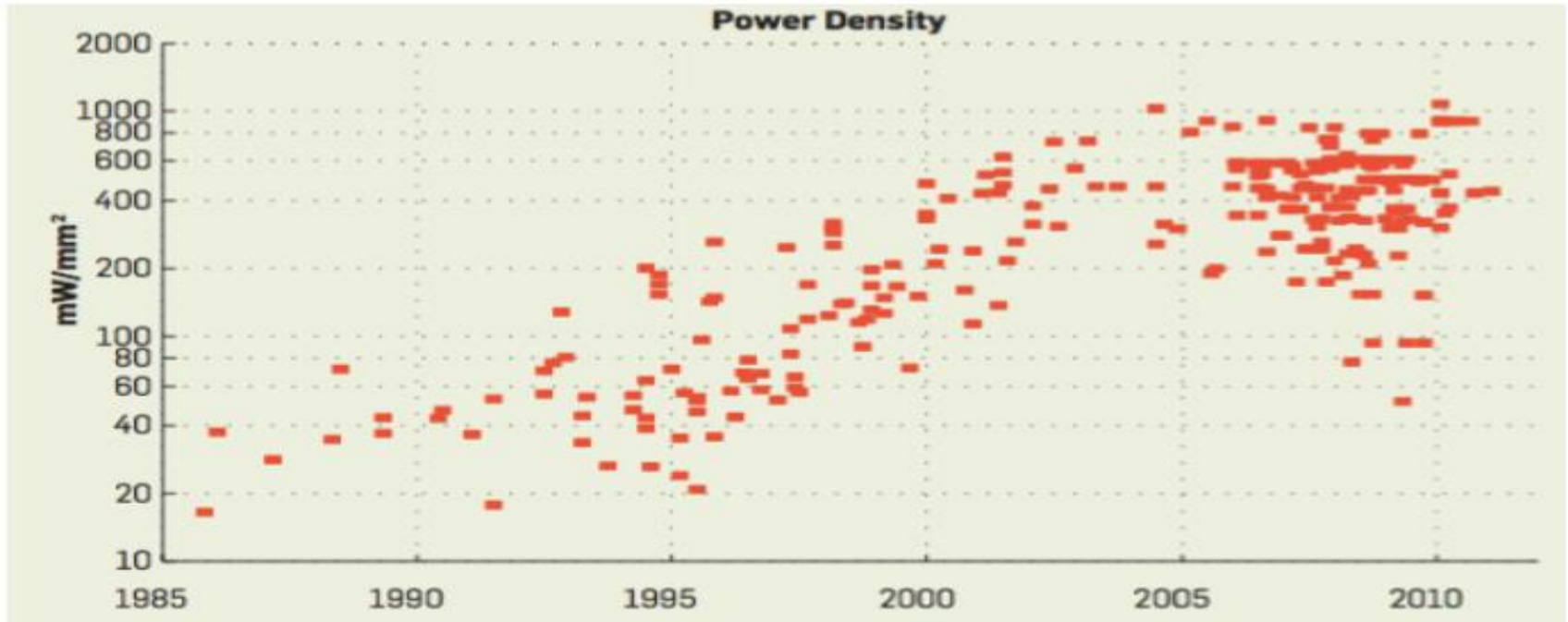
Impact of VLSI in processor trends



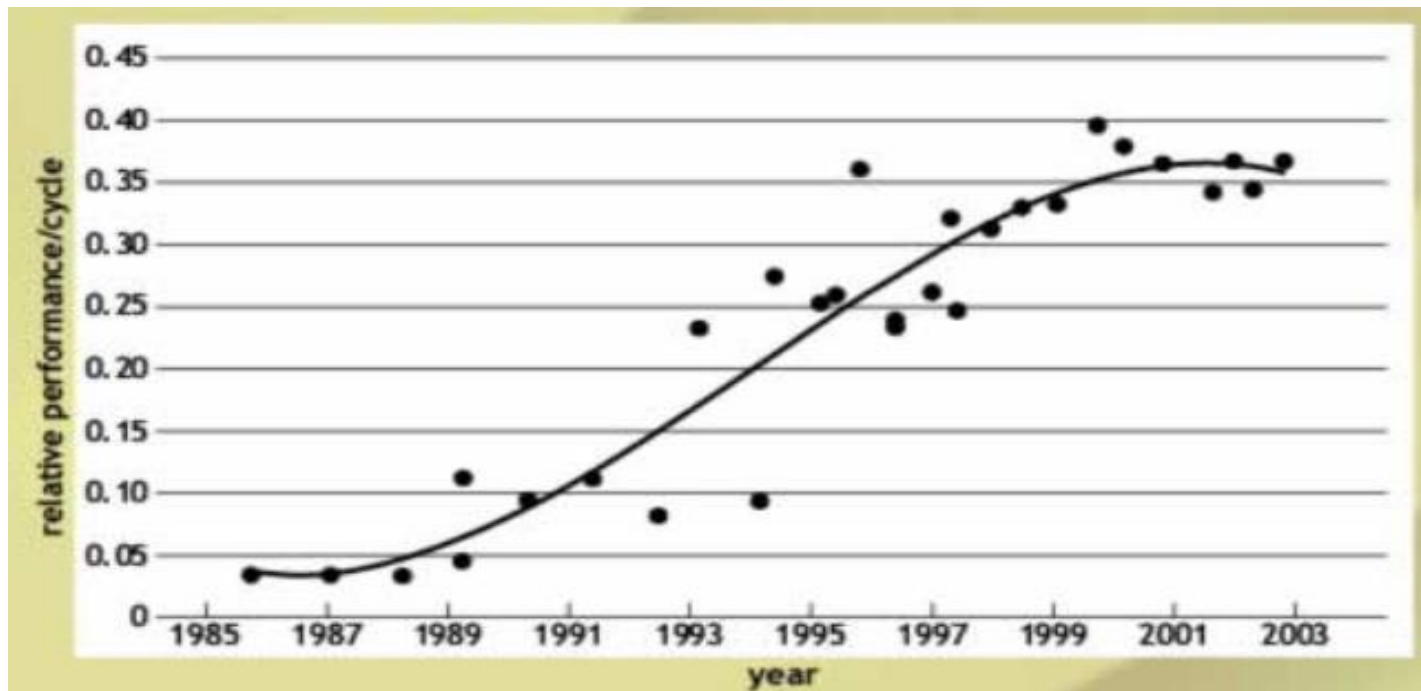
Frequency Wall



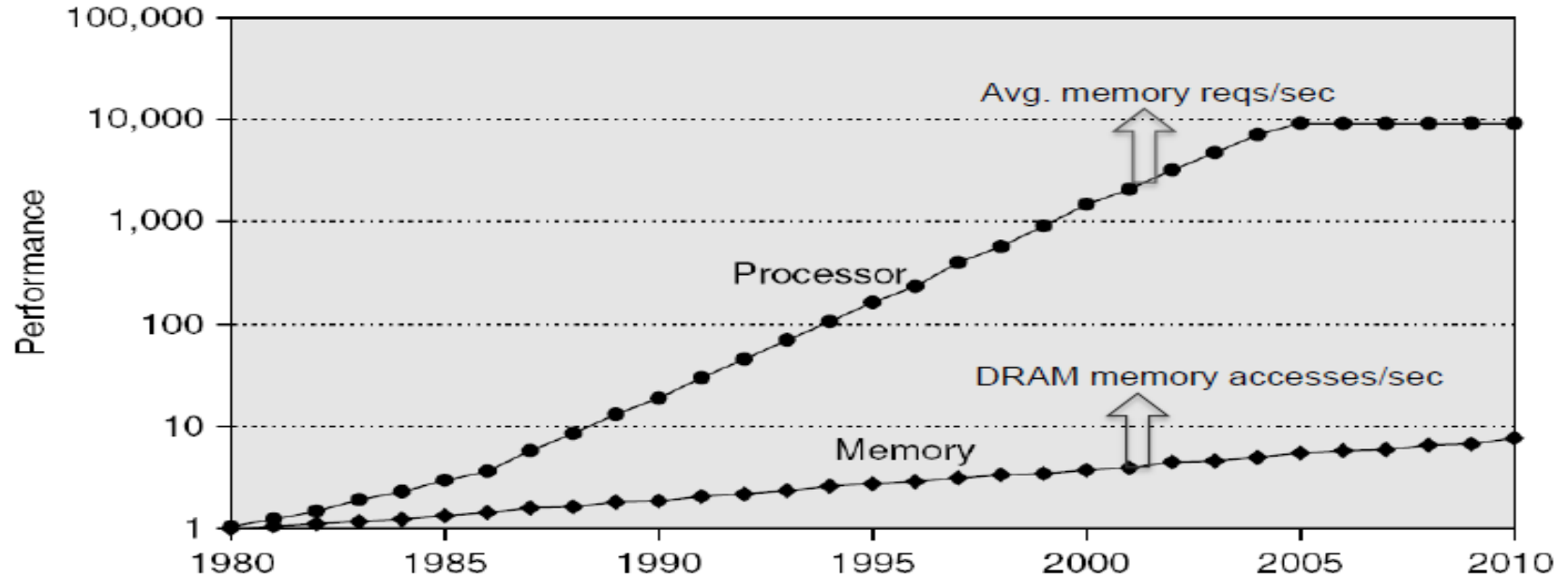
Power Wall



ILP Wall

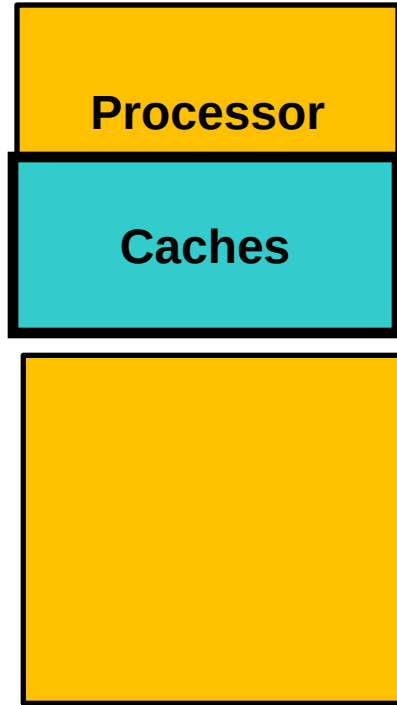


Memory Wall



Paradigm Shift to Multicore

2004 SoC : 90 nm

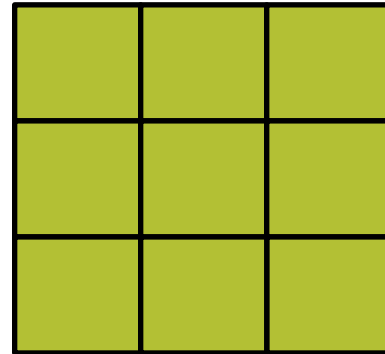
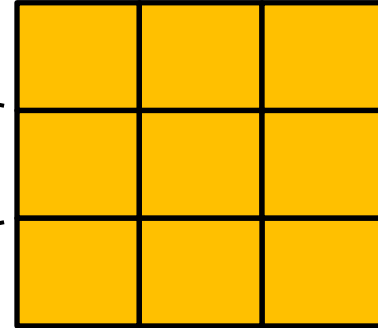


P
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2017 SoC : 10 nm



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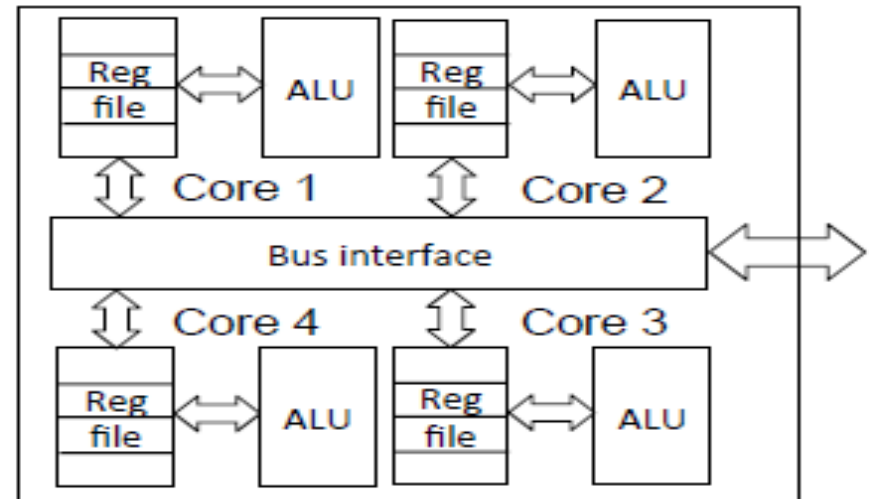
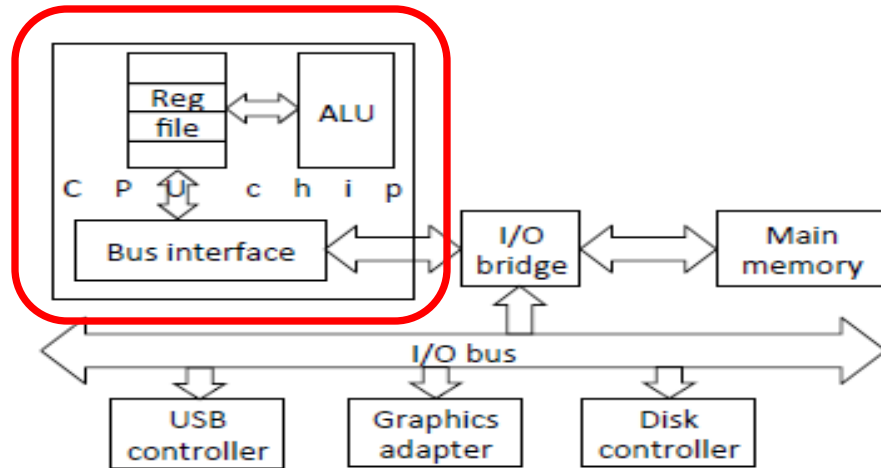
Multiple slower processors vs single fast powerful processor

Paradigm Shift to Multicore

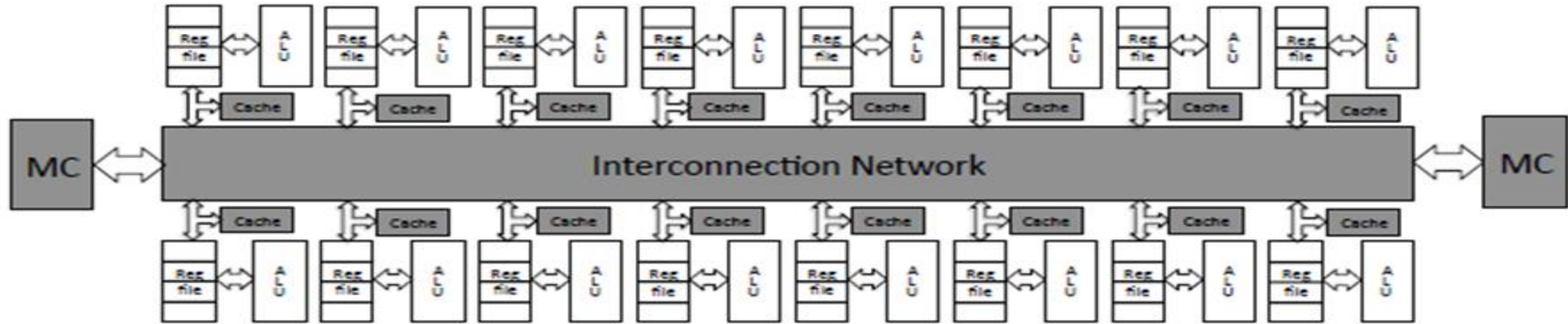


Multiple slower processors is better than single fast powerful processor

What is Multicore ?



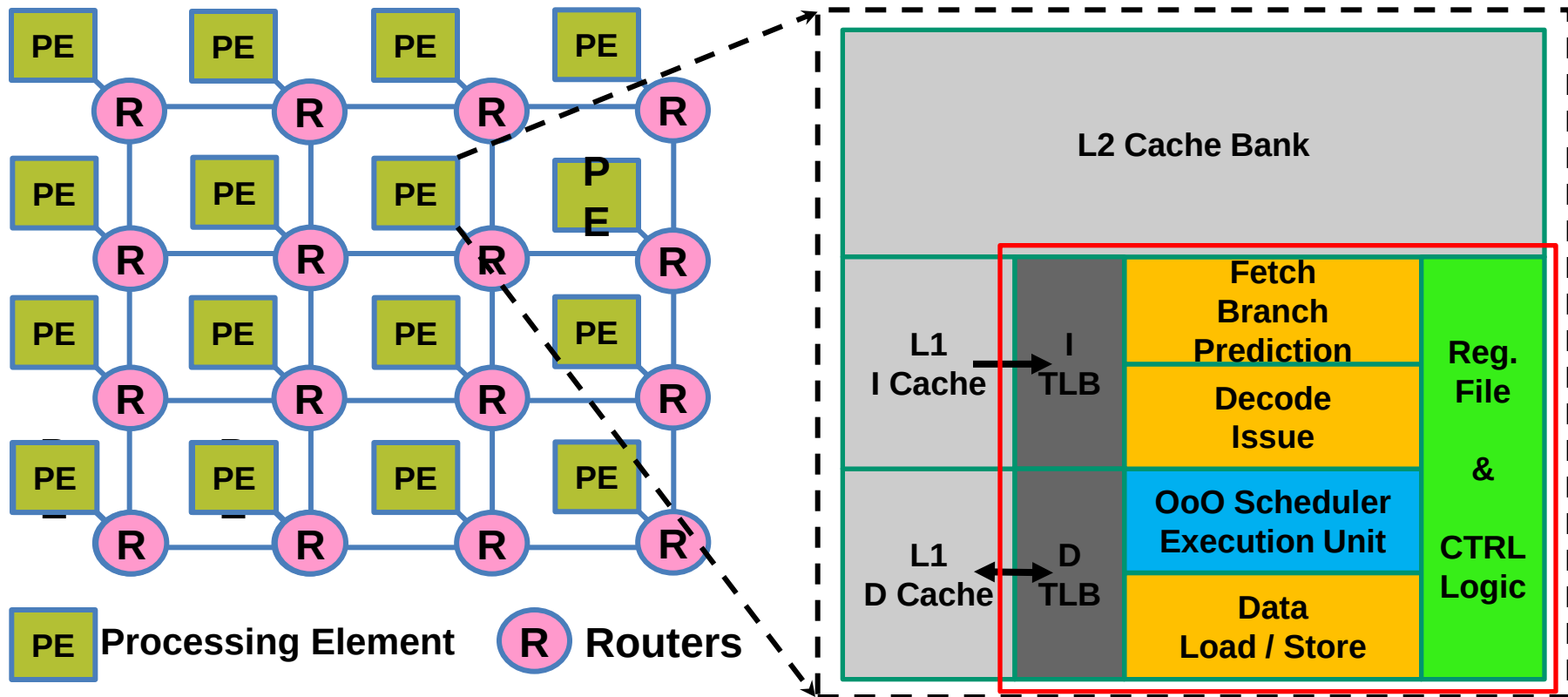
What is Multicore ?



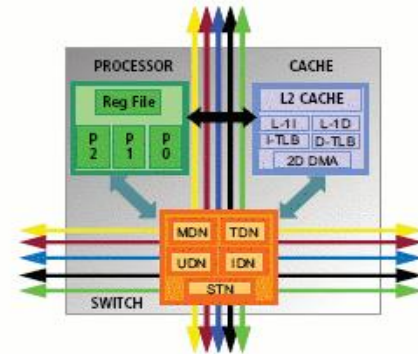
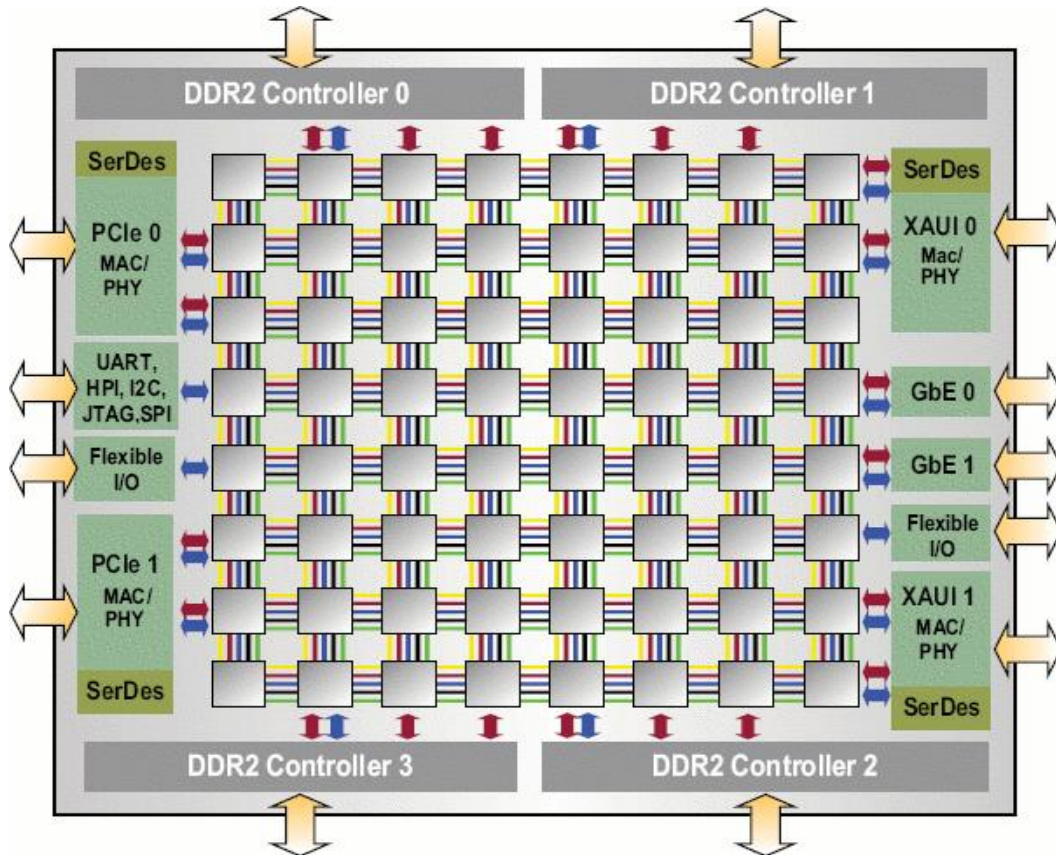
❖ How these cores communicate ?

❖ What is the best interconnection mechanism ?

Tiled Chip Many-Core Processor (TCMP)

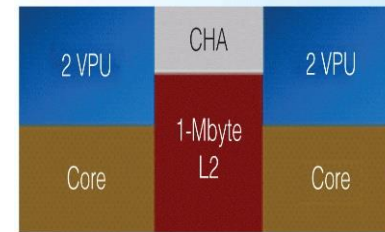
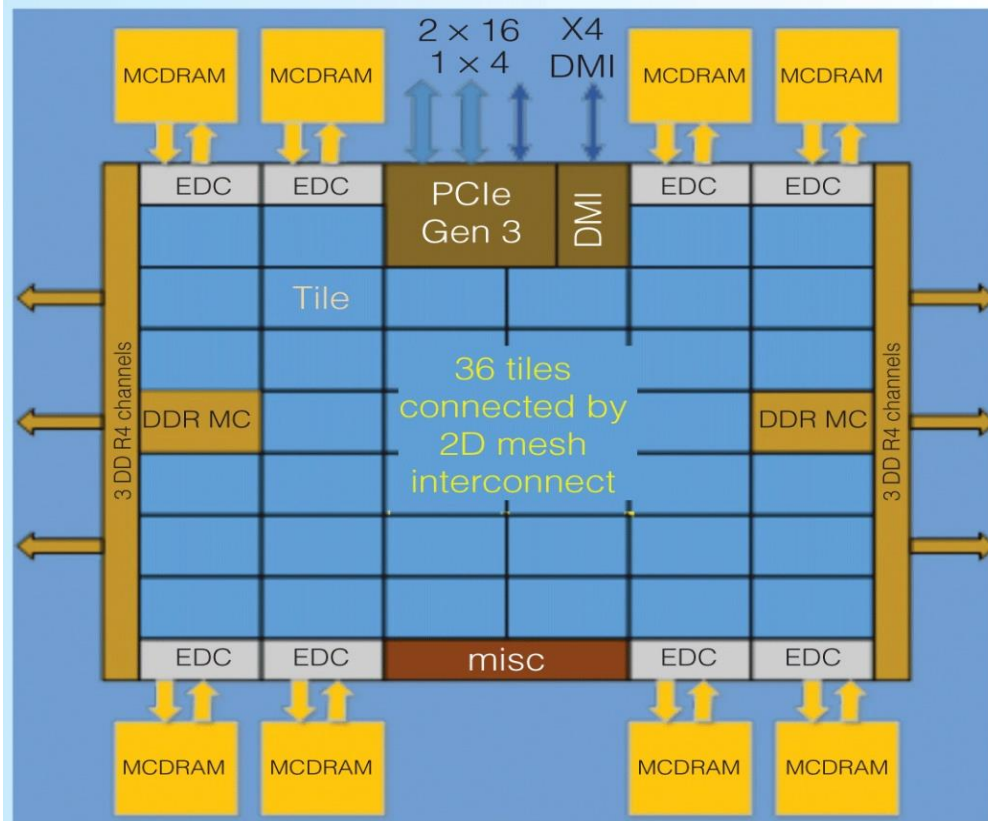


State-of-the-Art Architectures



Tiler64

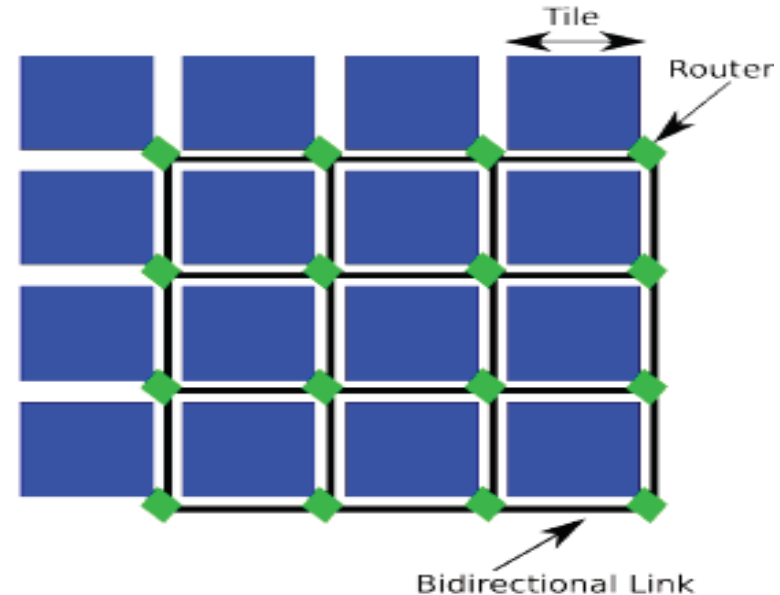
State-of-the-Art Architectures



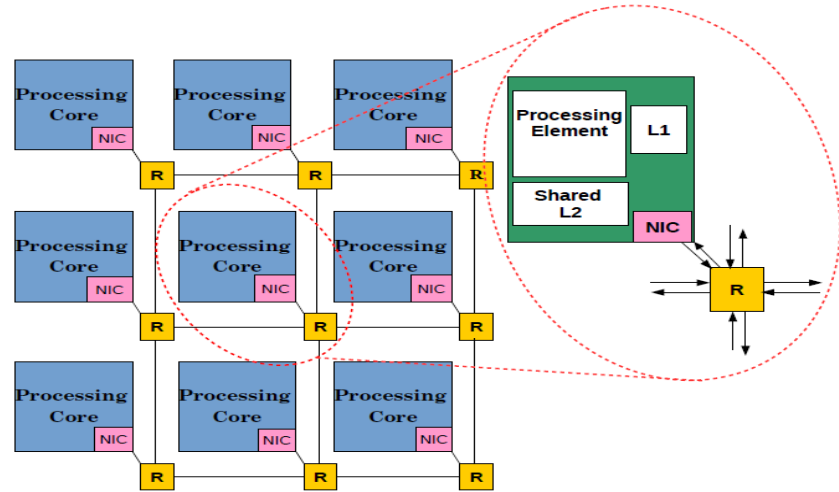
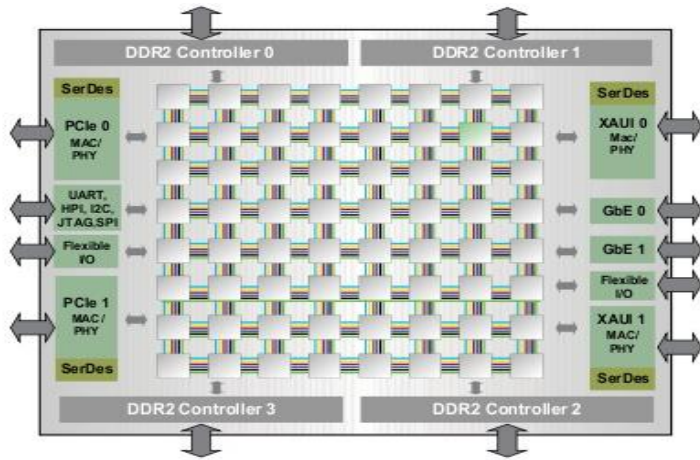
Intel KNL

What is NoC ?

- ❖ Processing units interconnected via packet based network
- ❖ Each resource is called as a **'tile'**
- ❖ All resources organized as rectangular tiles on the chip.
- ❖ Each tile have an address - (X, Y)
- ❖ Tiles interconnected by network of routers
- ❖ Communication by packet transmission

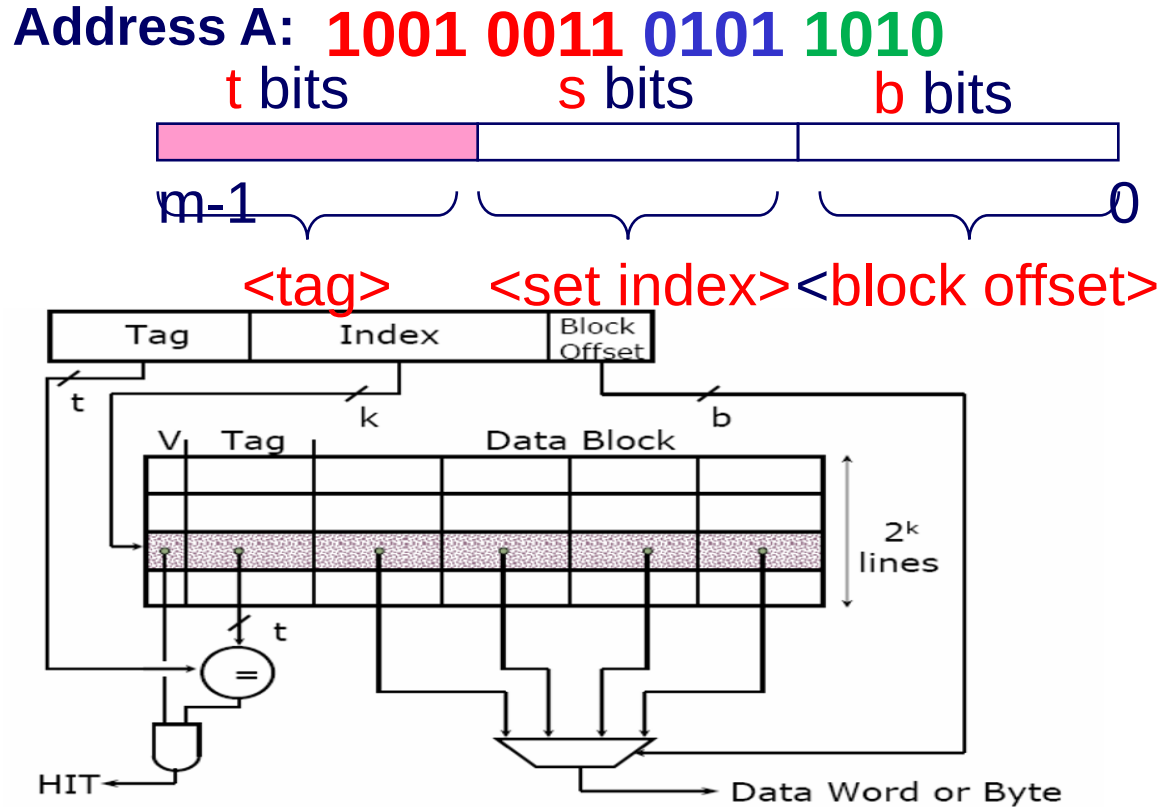
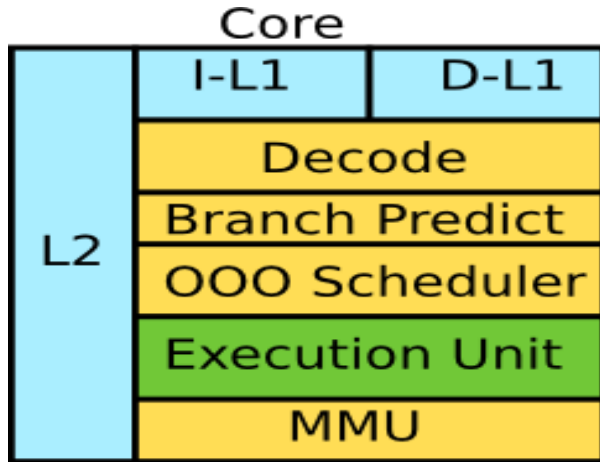


Routers and Tiles

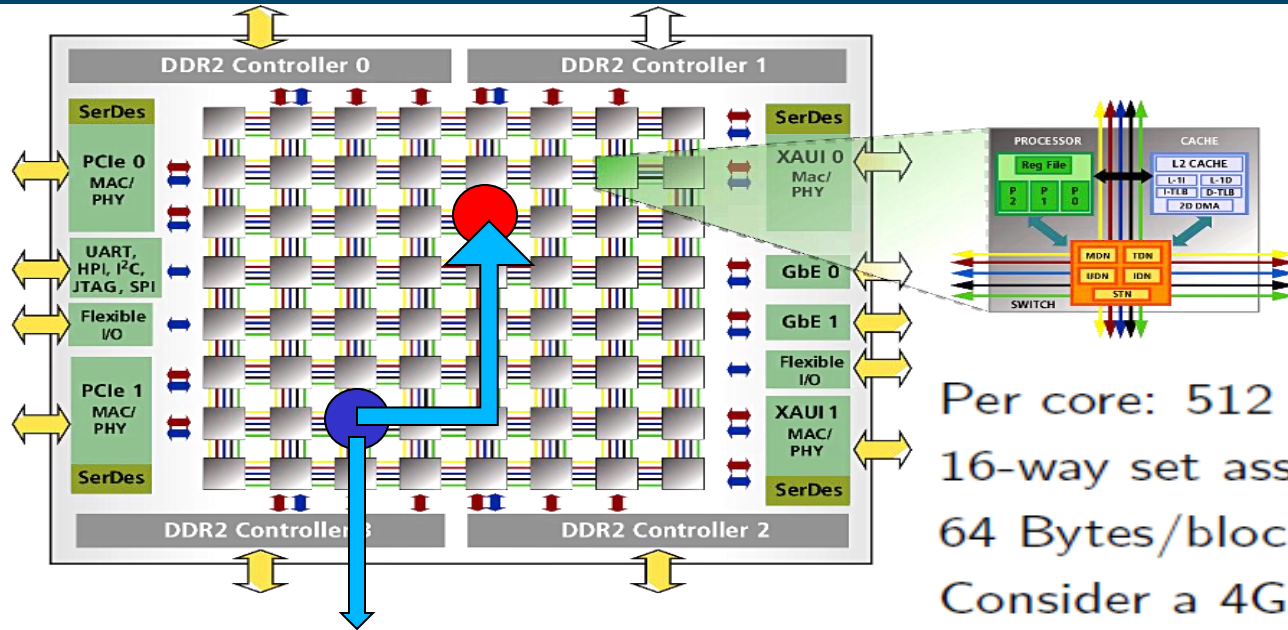


- ❖ East, West, North and South neighbors
- ❖ Packets are divided into flow control units called flits
- ❖ L1 and L2 cache misses create NoC traffic packets

What is the role of on-chip cache ?



On-Chip Cache Address Mapping



Per core: 512 KB

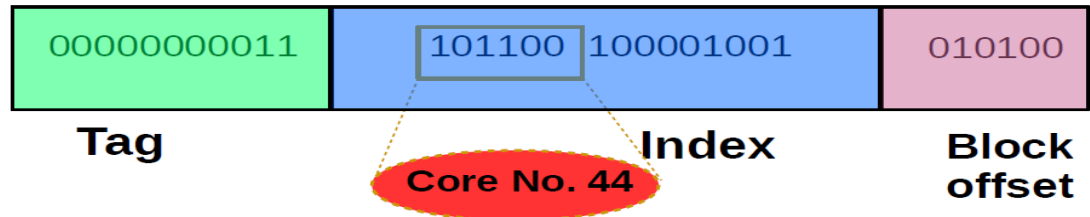
16-way set associative mapping

64 Bytes/block

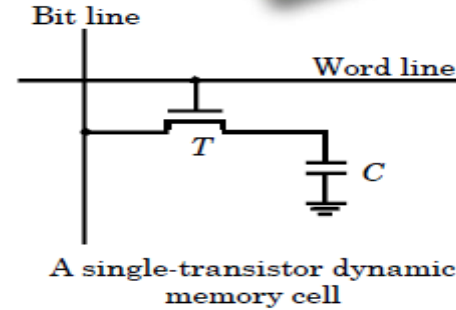
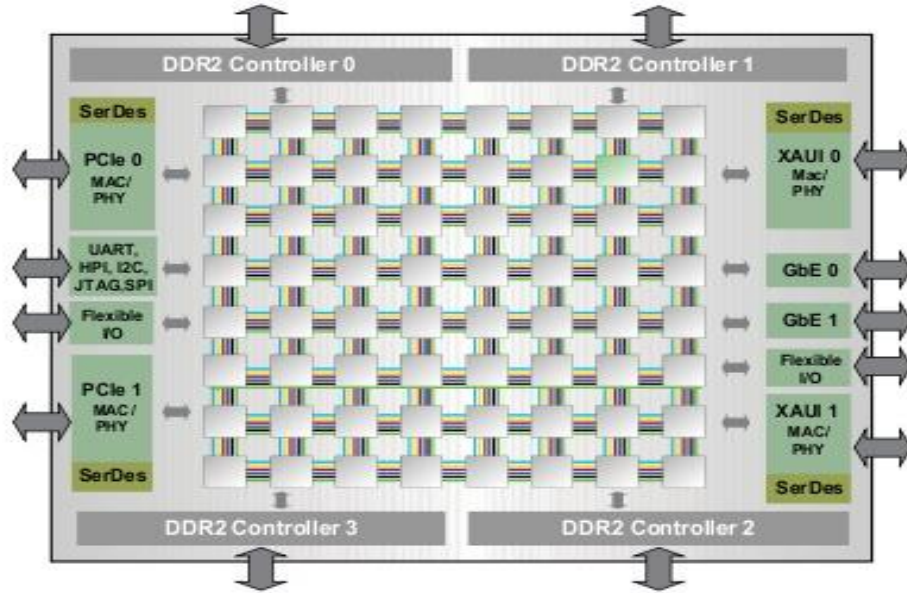
Consider a 4GB RAM (2^{32} bits)

Hex Address : 0x764254

11 bits	15 bits	6 bits
Tag	Index	Block offset



What is the role of memory controllers?



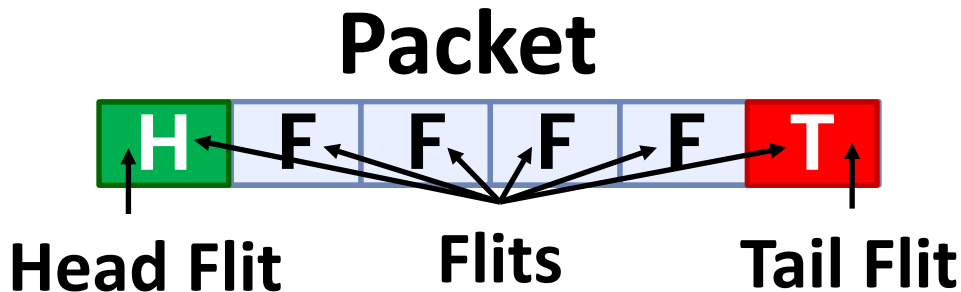
Packets & Flits

❖ Packet

- ❖ Unit of transfer for network

❖ Flit

- ❖ Basic unit of transfer between a pair of routers
- ❖ Unit of flow control within network





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